Assessment of Drinking Water Quality Using Water Quality Index: A Review

Article *in* Water Conservation Science and Engineering · January 2023

DOI: 10.1007/s41101-023-00185-0

2 authors:

Atanu Manna
Vidyasagar University

Debasish Biswas
Vidyasagar University

Water Conservation Science and Engineering (2023) 8:6
https://doi.org/10.1007/s41101-023-00185-0

REVIEW PAPER

Assessment of Drinking Water Quality Using Water Quality Index: A Review

Atanu Manna[1] · **Debasish Biswas**[2]

Received: 22 September 2022 / Revised: 8 January 2023 / Accepted: 22 January 2023

Abstract
Nowadays, declining water quality is a significant concern for the world because of rapid population growth, agricultural and industrial activity enhancement, global warming, and climate change influencing hydrological cycles. Assessing water quality becomes necessary by using a suitable method to reduce the risk of geochemical contaminants. Water's physical and chemical properties are compared to a standard guideline to determine its quality. The water quality index (WQI) model is a commonly helpful technique for evaluating surface and groundwater quality. The model mainly employs aggregation techniques to diminish large amounts of data to a sole value. The WQI model has been used across the globe to assess ground and surface water using regional standards. The model has become popular for its ease of use and general structure. Typically, WQI models include five stages: (1) choosing water quality indicators, (2) generating sub-parameters for each variable, (3) calculating variable weighting numbers, (4) aggregating sub-parameters to finding the total WQI value, and (5) classification of WQI value to highlight the category of water quality. In addition, the model creates ambiguity when converting vast volumes of data into a single value. The study considered 2011–2021 blinded peer-reviewed articles and book chapters to assess WQI models and their application in evaluating drinking water quality. This study mainly concentrated on the comparison of WQI models and their applications. The study also focused on the selection of parameters and problems associated with the accuracy of the models.

Keywords Water quality index · Water quality parameters · Model accuracy · Aggregation method

Introduction

Drinking water deterioration is becoming a global concern due to widespread human activities [1]. However, humans and nature are responsible for the decline of surface and ground water quality (WQ) [2]. Natural factors like lithological, topographical, climatic, atmospheric, and hydrological factors are the principal variables that influence WQ [3]. In addition, waste disposal, production, livestock farming, and mining are the primary variables affecting WQ [1].

✉ Debasish Biswas
debasish762022@gmail.com

Atanu Manna
atanumanna329@gmail.com

[1] Centre for Environmental Studies, Vidyasagar University, Midnapore 721102, India

[2] Department of Business Administration, Vidyasagar University, Midnapore, West Bengal 721102, India

Nowadays, most developing countries face enormous problems maintaining WQ while emphasising water supply and hygiene [2, 4, 5]. On the other hand, it is also noticeable that developed countries are combating to enhance and manage the proper water quality to maintain increasing population pressure and public health [6, 7]. Due to widespread groundwater contamination, Drinking Water Quality (DWQ) management has received much attention in the previous decade [8–11]. WQ management and control include the gathering and statistical analysis of large data sets on WQ to assess water attributes [12–16]. Several techniques have been developed for assessing the drinking WQ; the WQI is one such method for evaluating the WQ [17–19]. WQI models are centred on the aggregation function that helps to analyse vast spatio-temporal datasets to generate a sole value to determine the category of WQ [20–22]. In addition, the index provides a sole number for the aquatic environment that contains the WQ-related information that is easier for the utility authorities to understand [2, 23–25].

Published online: 30 January 2023

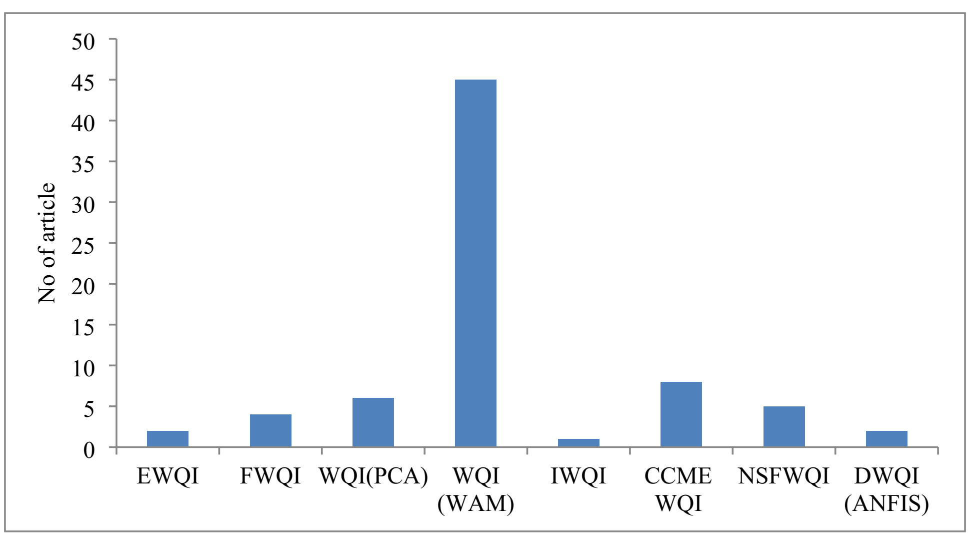
A WQI principally contains five stages to assess DWQ. At first, WQ parameter selection is necessary. Secondly, a dimensionless single value-based sub-index is generated from the WQ parameters. Third, the relative weighting factor is calculated based on the guidelines or mathematical functions. Fourth, the sole WQI value is created based on the weighting factors and sub-index value using the aggregation function [2, 26–32]. At last, classification is done based on the WQI value. However, there are several WQI models developed to measure DWQ. The models differ mainly in aggregation techniques, parameter selection process, weighting factor, and model structure to best combine the model in different locations [2, 15, 33]. It has been noted that the WQI components of the model are chosen based on WHO or country-based standard guidelines such as BIS for India [34, 35] or expert judgements [20, 36]. It is also worth noting that some models are area-specific to decrease model uncertainty like the WQI model has four levels with varying degrees of uncertainty [2, 15].

The principal aim of this study is to review the WQI models from the perspective of drinking water quality assessment. The study also deliberately explores the eclipsing and uncertainty concern associated with the WQI models. The researcher reviewed a total of 73 recently published blinded peer-reviewed articles (2011–2022) assessing drinking water quality using WQI models in this study. Among 73 articles, the researcher identified eight WQI models extensively applied to evaluate DWQ. In the second section, the chapter showcases the evolution of WQI models and their development. The third section includes a general overview of the five stages of WQI models for a weighted index such as (1) WQ indicators selection, (2) generating sub-parameters for each variable, (3) components weighting, (4) WQI value generation based on the aggregation function, and (5) WQI classification. However, the stages will be four for the unweighted method. The fourth section briefly discusses the eight WQI models used in drinking water quality assessment. The fifth section highlights the primary findings of the review. At last, the sixth section describes the conclusions of the study.

Evolution of Drinking WQI Models

The evolution of WQI models started over fifty years ago [15]. In the 1960s, Horton first developed the WQI model based on ten WQ indicators to highlight the WQ of a water body [37]. In 1965, the US National Sanitation Foundation (NSF) developed the WQI model based on biological, chemical, and physical parameters using the Delphi method. NSFWQI primarily includes multiplicative and additive functions to measure the WQ [38, 39]. The CCME, on the other hand, used a different method, such as the multivariate analytical technique, in aggregating the measured parameter values [37]. As a result, CCME WQI has become the most accepted measure of DWQ and is also applied to measure global drinking water quality index (GDWQI) [2, 40]. It is observed that most of the models use some common indicators to measure WQI, such as pH, TDS, nitrates, turbidity, DO, and different metal components [1, 2, 41]. With the advancement of time, the WQI models evolved into measuring WQ more accurately, and refined models were also observed with time [42]. More than 35 WQI models were followed and used by different countries for their surface and groundwater quality assessment [2]. However, all models have some advantages and drawbacks over other models, which is still challenging for researchers. In this study, the researcher observed eight WQI models, the most modified versions of the ancient models. It is observed that most of the WQI models were used to assess bore well/dug well-based groundwater quality assessment, as shown in Fig. 1. It is also noticeable that most researchers adopted a weighted arithmetic method (WAM)-based WQI model to assess drinking water suitability. However,

Fig. 1 Drinking WQI application summary (2011–2021). Source: The Author



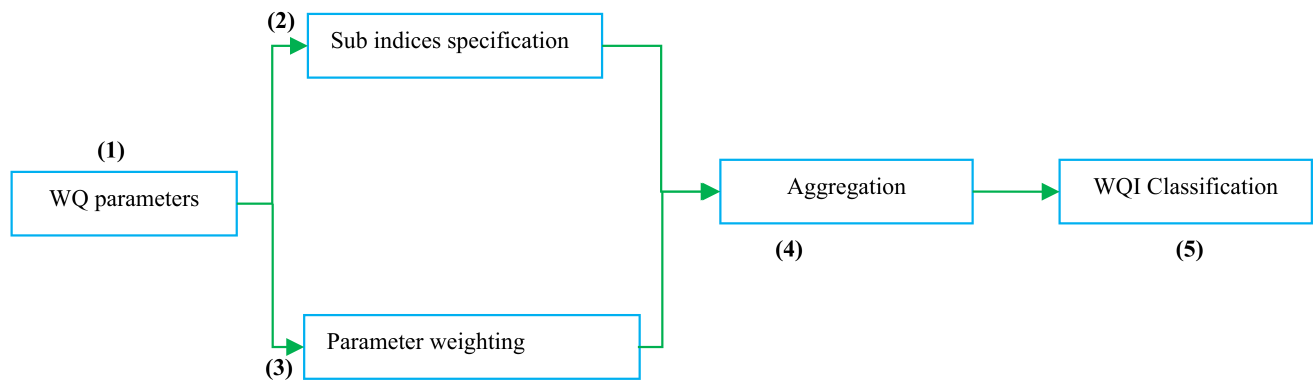


Fig. 2 Five stages of WQI model. Source: The Author

several Multi-criteria-decision-making (MCDM) developed in recent years to improve the measurement technique and reduce uncertainties, such as compromise programming (CP), data envelopment analysis (DEA), analytical hierarchical process (AHP), and analytical network process (ANP) [37].

WQI Model Construction (5 Stages)

Drinking water quality index (WQI) models primarily consist of five stages, and an overview of the structure is highlighted in Fig. 2 [2, 43]. The five stages are:

1) WQ indicators selection: one or more WQ elements are chosen to include for water quality evaluation.
2) Generating sub-parameters for each variable: all the sub-parameter values are converted into sub-indices (unit less number) with the help of parameter concentration.
3) Parameter weighting: Parameter weights are provided based on the importance of the parameter in the water quality assessment.
4) Computation of WQI value: In this stage, the aggregation function determines the sole WQI value. Stage 2 (sub-indices) and stage 3 (weighting values) are combined to get a single WQI value.

Table 1 summary of WQI models structure

WQI model	References	Steps involved in the model			
		No. of parameters and selection procedure	Sub-indexing procedure	Parameter weighting	Aggregation techniques
EWQI	[4, 12, 45]	• 14 parameters suggested • Information entropy	• Utilised standard guideline of water quality	• Entropy weight of each parameter	• Utilises additive function
FWQI	[46]	• 10 parameters suggested • Standard procedure (APHA)	• Utilised standard guideline of water quality	• Fixed and unequal weight	• Mamdani fuzzy model
WQI(PCA)	[47]	• 9 parameters suggested • Delphi method	• Parameters value directly used in sub-index	• Weighting factor • Sum of weighting value is 25	• Additive function
WQI (weighted arithmatic method)	[14, 27, 35, 41]	• 13 to 15 parameters suggested • According to WQ guidelines (BIS or WHO)	• Adopted guideline value for creating sub-index • Quality rating scale	• Weighting factor based on the relative importance • Sum of relative weight value is 1	• Additive function
IWQI	[25]	• 9 parameters suggested • Based on WQ guidelines	• Based on the WQ guidelines (MPL)	• Not required	• Simple additive function
CCME WQI	[1, 48]	• 12 parameters suggested • Delphi method	• Not required	• Not required	• Used three-factor-based fixed functions
NSFWQI	[26]	• 11 parameters • Delphi method	• Using conversion curve (range from 0 to 100) • Expert opinion	• Weighting factor based on the relative importance	• Additive mathematical function
DWQI (ANFIS)	[28]	• 18 parameters suggested	• Relative weight and concentration-based rating	• Relative weight based on the importance	• Simple additive aggregation function • Artificial neural network

Table 2 WQ parameters involved in WQI models

WQI			EWQI	FWQI	WQI (PCA)	WQI (WAM)	IWQI	CCME WQI	NSFWQI	DWQI (ANFIS)
Common WQ parameters	Physical	Temperature	✓						✓	✓
		Colour			✓					
		Conductivity			✓	✓		✓		✓
		SS			✓					
		Total hardness			✓	✓	✓			
	Chemical	PH	✓	✓	✓	✓	✓	✓	✓	✓
		TDS		✓			✓		✓	✓
		DO			✓				✓	
		COD			✓					
		Aluminium			✓					
		SO4	✓	✓	✓	✓	✓	✓		✓
		Cl	✓	✓	✓	✓	✓	✓		✓
		NO3	✓	✓	✓	✓	✓	✓		✓
	Biological	F. Coliforms			✓				✓	
		T. Coliforms			✓					
Additional parameters		Na	✓	✓		✓	✓	✓		✓
		K+	✓	✓		✓	✓			✓
		Ca	✓	✓		✓	✓	✓		✓
		Mg	✓	✓		✓	✓	✓		✓
		PO_4	✓						✓	
		HCO_3	✓	✓		✓		✓		✓
Toxics, pesticides, and trace metal		Cd			✓					
		Mn	✓		✓					
		Zn	✓		✓					
		Cr			✓					
		Cu			✓					
		Pb	✓							
		Fe	✓		✓					
		Ba			✓					
Others					Co, F, Hg, Ni, Se,			CO3	Turbidity, BOD	Turbidity, Alk, F

5) WQI value classification: The sole WQI value is classified based on the rating scale to determine the level of water quality for drinking.

A brief discussion has been made in the below section to highlight the stages of the WQI models and their importance.

Parameter Selection

Water quality components identification and selection is the first and critical stage of the drinking WQI process. However, several variations were observed in WQI parameter selection and the reasons for parameter selection, such as data availability, expert opinion, and environmental significance [14, 22, 44]. Table 1 briefly discusses the parameters included in the WQI models. It is observed that some parameters are included commonly in most of the models, such as total hardness (TH), conductivity, pH, SO4, Cl, No3, Na, K, Ca, and Mg (see Table 2). It is also noticed that several parameters vary model by model, as shown in Table 2 [2, 23].

WQI model components were primarily chosen depending on the data availability, the opinion of an expert, and the importance of WQ components [2, 13, 38, 39]. Numerous scholars have modified the WQ parameters depending on the data accessibility and availability [49]. Multiple WQI models did not consider several parameters such as toxic compounds, microbiological components, and suspended solids due to a lack of modern laboratory and high analytical cost [2, 15, 46, 48]. It is observed that the parameter selection varies depending on the type of study, such as Environmental Impact Assessment, Contamination, Pollution, and Assessment of DWQ [50]. Most of the WQI models use the Delphi function for selecting WQ components. The significant components are chosen based on expert opinions [2]. It is noticed that there are no specific guidelines or regulations for selecting the WQ components to include in the model. However, there are three principal systems observed concerning parameter selection [37, 51] and those are as follows:

(i) Fixed system: In this system, WQI indices are considered some set of predefined parameters that the WQI user has chosen as the best suitable collection of variables to gauge the value of the final index and new parameters are not included in this model, and rigidity emerges. The scholars use only those verified parameters to check the water quality in different regions [37, 51, 52]. However, a fixed set of criteria presents serious issues, such as challenges in drawing comparisons between water supplies and monitored places [52].

(ii) Open system: In this system, some WQI model allows a minimal set of parameters based on their characteristics and impact on water resources. Although the WQIs are adaptable and less restrictive, there is a severe problem when comparing the findings from different monitoring sites. As a result, these WQIs may differ from the area as the parameters vary regionally. The computation of the final index does not include a maximum parameter selection number. The system allows users to adopt numerous parameters from the standard guideline and helps avoid the rigidity concern.

(iii) Mixed system: the mixed system includes open and fixed systems and considers both primary and additional components in the WQI model. The fundamental and extra parameters make up the mixed system. Only if one of the extra parameters has a higher sub-index value than the aggregated index value based on the fundamental parameters are additional parameters considered in the final index computation. In this situation, the final aggregated index value needs to be revised to include or consider any new parameters with higher sub-index values [51].

Most scholars are highlighting expert opinion and statistical method-based approaches for selecting the best parameters for the study. Most scholars have recently been using statistical methods to determine the number of parameter selections [53–55]. According to M.G. Uddin et al. [55–57], there are mainly three groups that help in feature selection, and those are (1) wrapper method which allows unsupervised and supervised classification algorithms (such as Random Forest, Decision tree, Boruta, and XGBoost to extract best parameters to enhance the model accuracy and reliability; (2 filter method that allows the rank based statistical technique to find the best suitable parameters such as correlation, MI, PCA, Chi2 test, and Relief technique; and (3 embedded technique which combined wrapper and filter both method to reduce the model error and improve model accuracy.

Sub-index Specification

The sub-indexing process is the crucial second stage in the WQI model, which helps to alter the concentration of parameters into a unitless value [58]. Numerous drinking WQI models have established sub-indices rooted in the traditional WQ guideline values provided by the authorities, such as WHO and BIS [12, 46]. However, the CCME WQI model is one of the WQI models investigated that skips this phase and uses parameter concentration to execute the final aggregate function without calculating sub-indices [1, 2, 48, 59]. Sub-indexing procedures for the selected models are highlighted in Table 1.

a) Parameter concentrations

Sub-index (unitless) generation based on the parameter concentration technique is one of the most straightforward techniques, and most of the drinking WQI models used this technique to convert parameters value into unitless sub-indices [38]. Horton first used the parameter concentration technique in his WQI sub-indices calculation [60, 61]. Most models use the variable intensities value as the sub-indices value without modification [2, 20, 62].

b) Linear interpolated functions

NSF WQI specifically applied variable stretches from the standard guideline of WQ to measure the sub-index values linearly [2]. In this model, the sub-index spans from 0 to 100. Therefore, 0 was allocated if the parameter concentration value exceeded the standard guideline value. On the other hand, if the parameter concentration did not achieve the guideline mark, then the sub-index value was consigned to 100 [26, 63–65].

Recently, it has been noticed that the sub-index stage is also involved with eclipsing and uncertainty issues. However, there are three popular methods are considered in the sub-indexing process, and those are:

(i) Expert judgement: This is one of the crucial techniques used in the case of sub-index specification. Three techniques—individual interviews, virtual focus groups, and the Delphi method—are often used to include parameter selection into the expert judgement process. Typically, the Delphi approach is used by the WQI expert group to gather pertinent data to develop sub-index functions using surveys [51].
(ii) Statistical methods: Some statistical techniques are applied in sub-index specification, such as Principal Component Analysis (PCA), Principal Financial Analysis (PFA), cluster analysis, and Pearson correlation [51]. However, machine learning techniques have recently become familiar in parameter selection and specification [55]. These techniques help reduce uncertainty concerns.
(iii) Use of water quality standards: Based on the allowable limitations from the guideline standards, such as WHO standards, national guidelines, or international directives, is another method to build rating curves or sub-index functions [37, 51].

Parameter Weighting

Generally, the weight value of WQ items is established based on the respective magnitude of WQ characteristics or by using standard WQ recommendations [33]. Most models use relative weighting methods such as WAM-based WQI models [27] and PCA-based WQI models [47] to calculate parameter weight where the sum of the relative weights was equivalent to 1 [66]. In this study, CCME and IWQI models do not use the relative weight method for weighting the parameter to measure WQ. The parameter weighting step has a crucial influence on the WQI value. Applying the unequal weighting method is helpful for the WQI models to reduce uncertainty [67]. In other words, if any model provides an inappropriate weighting method, it will significantly affect the WQ assessment. Several WQI models, on the other hand, use expert views to inform the component weighting process [2, 66].

This phase principally deals with equal and unequal weights concerning parameter weighting. In the case of equal weighting, expert judgement plays a crucial role in parameter weighting [37]. However, assigning different weights for the same parameter by the expert panel may lead the subjectivity concern for the final index [51]. However, for the drinking water quality assessment, the scholar should focus on the most weighted parameter as it is susceptible to the final index. In the case of unequal weighting, two methods are widely applicable, and those are the Delphi method and analytical hierarchy process (AHP). Some other participatory-based methods, such as the improved Simos' technique and the budget allocation procedure (BAP), have been utilised [51]. The Delphi method mainly provides the average value of the experts to create parameter weights. On the other hand, AHP is also a technique used to get expert opinions for deciding the parameter weights. In this technique, pair-wise comparison metrics have been made for evaluating parameter weights. However, in recent decades, some scholars used the AHP technique to determine the parameter weights for machine learning-based water quality evaluation, such as artificial neural network (ANN), support vector machine (SVM), and random forest (RF) [55].

Aggregating Function

The fourth stage of the WQI model is aggregating function. Its primary purpose is to combine the variable sub-indexes into a sole WQI score [42, 68, 69]. This single value is then categorised for better understanding. However, most models use additive or multiplicative formulas or an amalgamation of these two functions to calculate the WQI value (see Table 1) [2]. However, [53, 54] include eight different aggregation functions in their study and highlight that every model has uncertainty. They highlight that the machine learning-based approach is less uncertainty concern than the traditional WQI models. A brief discussion has been made below concerning different types of aggregation functions.

a) Additive functions

In this study, the researcher observed that most scholars applied additive functions to assess drinking water quality (DWQ). The additive function is as follows:

$$\mathrm{WQI} = \sum_{i=1}^{n} s_i w_i$$

where S_i is the sub-index score for the i element; n is the total no. of components; w_i is the corresponding component weight score.

b) Multiplicative functions

Some WQI models applied multiplicative functions to measure WQI for assessing drinking water quality, such as the NSF WQI model. The multiplicative aggregation function is as follows:

$$\mathrm{WQI} = \prod_{i=1}^{n} s_i^{w_i}$$

c) Combined aggregating functions

This method mainly combined the additive and multiplicative functions. NSF index uses both multiplicative and additive functions [2, 53, 54].

d) Root mean square function

M. G. Uddin et al. [53, 54] proposed a root mean square (RMS) function (unweighted) to minimise the aggregation concern. The function is as follows:

$$\mathrm{RMS} = \sqrt{\frac{1}{n}\sum_{i-1}^{n} s_i^2}$$

e) Minimum operator function

Smith [70] developed this technique and measured New Zeeland's stream and river water quality. Later, Shah & Joshi [71] also applied this technique to measure surface water quality in India. The formula is as follows:

$$\mathrm{WQI} = \mathrm{Min}(S_i + S_{i+1} + S_{i+2} \ldots \ldots ..\mathrm{Isubn}$$

f) Non-linear aggregation function

This technique mainly used a unique logarithm function [72], and it is as follows:

$$\mathrm{WQI} = \log\left[\frac{(DO)^{1.5}}{(3.8)^{TP}(\mathrm{Turb}i)^{0.15}(15)^{\mathrm{fecal}/10000} + 0.14SC^{0.5}}\right]$$

g) Arithmetic mean function

This technique is developed by M. G. Uddin et al. [53, 54] to measure unweighted-based WQI value. The function is as follows:

$$\mathrm{AM} = \frac{1}{n}\sum\nolimits_{i=1}^{n} Si$$

Classification

This last stage of WQI mainly deals with classifying the WQI values for assessing water quality. It is noticed that most of the WQI methods use a rating scale for grading the water quality to highlight its suitability for drinking. However, a significant difference is observed in the classification technique based on the model structure [57]. The study provides a brief about the classes used by the scholars for different selected WQI models in the below section.

Primary WQI Models

The researcher identified eight principal WQI models used in the 24 articles to analyse drinking water suitability in this study. The summary of the model features is showcased in Table 1. It is observed that most scholars are used the WAM technique in WQI calculation, and the WQI (WAM) model accounts for more than 50% of the total models used in drinking water quality assessment (see Fig. 1). It is noticed that Mukate et al. developed new integrated WQI models in 2019, as it is the most recently published new model, and it helps to minimise the uncertainty of earlier WQI models. In this section, a brief discussion has been made concerning the eight models used in the DWQ assessment. The discussion will help us understand specific model structures such as parameter inclusion, sub-indexing process, parameter weighting process, and WQI measurement technique (see Table 1).

Entropy Water Quality Index

Many scholars in many nations use the entropy water quality index (EWQI) model to assess drinking suitability (see Fig. 1). The index has five processes for calculating WQI: parameter selection, sub-index measurement, weighting parameters, aggregating technique, and classification [4, 12, 45, 73].

1) Parameter selection

The EWQI model includes fourteen physical and chemical elements like pH, TDS, Cl, Ca, Mg, and others [12]. The model items are chosen based on the standard guidelines and expert opinions depending on the environmental importance. For example, Egbueri et al. [12] adopted pH and PO_4 outside the standard range.

2) Subindex specification

EWQI used the information entropy method to calculate sub-indices. Shannon introduced the Entropy measurement technique in 1948 [74]. The procedure helps to address the uncertainty concerning stochastic information [75]. The formula is as follows:

$$ei = -\frac{1}{In} n \sum_{i=1}^{z} PijInPij$$

where Pij is the probability of an event of the normalised value of the parameters j, and n is the no. of samples.

3) Parameter weighting

The EWQI model used the entropy weighting function to provide unequal weight to the parameters. The function is as follows:

$$wi = \frac{1 - ei}{\sum_{j=1}^{n}(1 - ei)}$$

4) Aggregation

In EWQI, the quality rating was measured before the aggregation function. Therefore, the formula below is used to compute the rating system for each variable.

$$qi = \frac{Ci}{Si} \times 100$$

where S_i denotes the WQ-recommended standard for each metric in water samples (NIS or WHO). C_i is the concentration of items in each water sample. The S_i value sometimes takes the upper limit when the standard value is in range.

At last, the EWQI value is calculated using the additive function. Then, the EWQI value is calculated based on the wj and qj values. The formula is as follows:

$$\text{EWQI} = \sum_{j=1}^{n} wi.qi$$

5) EWQI evaluation

EWQI model values are assessed based on the five ranks [12]. The ranks and their utility are as follows:

i) < 50 EWQI = Excellent
ii) 50–100 EWQI = Good
iii) 100–150 EWQI = Average
iv) 150–200 EWQI = Poor
v) > 200 EWQI = Extremely poor

FIS and AN-FIS WQI Model

Fuzzy interface systems (FIS) became known in the 1960s, and numerous scholars adopted this technique to assess environmental risk [2]. Recently, several scholars used the FIS technique to assess WQ [24, 76]. It is also noticed that most researchers have applied the FIS technique to evaluate the surface water quality [77–81]. However, few researchers applied FIS- and ANFIS-based WQI models to assess the DWQ [28, 46]. The FIS WQI model contains four stages, and those are as follows:

1) Parameter selection (Fuzzy set)

In the FIS WQI model, logical rules and set functions theory is applied to decide the WQ parameters but do not suggest any particular elements for assessing the WQ. Instead, the model includes correlation analysis to pick the model elements. Some researcher also allows expert opinion to select model parameters [2, 46, 82].

2) Sub-indexing (Fuzzy set operation procedure)

FIS technique helps to normalise the WQ parameters, and it provides numerical value as input and converts the values into a qualitative value with the help of FIS functions such as operators, member functions, sets, and fuzzy reasoning [2, 83].

3) Parameters weighting (Fuzzy logic function)

FIS logic function is applied to calculate the weight value of the selected parameters [2].

4) Aggregation

Different types of Fuzzy logic interface rules were involved in the WQ aggregation process, such as artificial neural FIS and Mamdani FIS [84]. At last, the defuzzification process is included to get the WQI value [36].

5) EQI evaluation

DWQI model's single values are classified into five categories. However, the class range is different in FIS and ANFIS models to assess WQ. The categories are as follows:

The classification for FIS-based WQI classes is Excellent (0–70), Good (30–120), Poor (70–230), Very poor (120–320), Unsuitable for drinking (230–500) [46]. The classification for AN-FIS-based WQI value classes are as follows: Excellent (< 50), Good (50–100), Poor (100–200), Very poor (200–300), Inappropriate (> 300) [28].

PCA and WAM-based WQI models

Most researchers have adopted the WQI model as established by Horton [2, 66]. The WQI model is modified several times depending on the region, parameter selection, parameter weighting, and aggregation function [33, 85]. Most of the academics in this inquiry employed the weighted arithmetic approach to determine DWQ [27, 35, 41, 86]. Different WQI models and their standard calculation process is highlighted in the below section.

1) Parameter selection

Several researchers applied different parameters to measure WQI based on the standard guidelines, expert opinions, regions, and data accessibility. In this study, we observed that Bouteraa et al. [27] used 11 parameters to measure WQI, Rabeiy [86] used 12 parameters to measure WQI, and Wagh et al. [41] used 13 parameters to measure WQI. Therefore, it can be said that there are no rules in the parameter selection. The most commonly used WQ parameters were pH, EC, Cl, Mn, Fe, SO4, HCO3, Mg, Ca, K, Na, and TH.

2) Sub-indexing

The researchers are using the standard guideline for calculating the sub-index, which is unitless [43]. However, some scholars are using PCA and factor analysis to minimise the number of dimensions [47].

3) Parameter weighting

To calculate the weighting factor, the model primarily employs an uneven weighting method, with the total values equal to one [2, 87]. Different types of unequal weight measurement techniques were also observed in this study. The equations are as follows:

The below formula is used by Alver [47], Rabeiy [86], Hui et al. [14], and Talat et al. [88]. The formula is:

$$Wi = \frac{wi}{\sum_{i=1}^{n} wi}$$

where w_i denotes the parameter's weight.

Another formula used by Bouteraa et al. [27]. That is:

$$Wi = K/S_i$$

where

$$K = \frac{1}{\sum_{i=1}^{n} \frac{1}{S_i}}$$

S_i = standard permissible value.

4) Aggregation

Before going to the WQI calculation, the quality rating was measured. The quality rating formula is as follows [27, 86]:

$$Qi = [(Vi - V0)/(Si - V0)$$

V_i is the ith parameter's measured value, and V_0 is the ith parameter's acceptable condition. Hui et al. [14] and Alver [47] used a different formula to measure quality rating (qi). The equation is:

$$qi = \frac{Ci}{Si} \times 100$$

C_i is the item's concentration, and S_i is the standard value of the ith parameter according to WQ guidelines.

Most of the scholars applied additive functions concerning the WQI measurement. The formula is as follows:

$$\mathrm{WQI} = \sum_{i=1}^{n} Wi \times qi$$

However, Bouteraa et al. [27] used different functions to measure WQI, and the equation is as follows:

$$\mathrm{WQI} = \sum_{i=1}^{n} QiWi / \sum_{i=1}^{n} Wi$$

5) WQI assessment

WQI values are categorised into five classes for better understanding, and the classes are as follows:

i) < 50 = Excellent
ii) 50–100 = Good
iii) 100–200 = Poor
iv) 200–300 = Very poor
v) > 300 = Unfit

NSF WQI Model

National Sanitation Foundation (NSF) WQI model was first introduced by Brown in 1965 [2, 89]. The model is a tweaked version of Horton's WQI, applied to assess surface WQ, such as rivers, lakes, and estuaries [2]. At present, some researchers adopted this model to evaluate WQ for drinking purposes (Table 1). The model includes four stages to measure WQ, like the Horton model.

1) Parameter selection

The NSF model used the Delphi method to select WQ parameters [2, 42]. As a result, the model mainly contains 11 parameters from five WQ groups: physical, chemical, microbiological, nutrients, and toxic. However, Al-Hamdany et al. [26] excluded some toxic parameters and used only nine WQ parameters to measure WQ.

2) Sub-indexing

The sub-indices calculation has been done using expert opinion. Sub-index values range from 0 to 1. When the measured values of the WQ parameter lie within the guideline range, provide 1; otherwise, the value will be 0. Some researchers calculated sub-indices using the conversion curve [2, 26].

3) Parameter weighting

The model adopted an unequal weighting approach to measuring the weights of the WQ items. As a result, the relative weight's sum becomes 1. However, the model includes expert opinion and environmental significance in measuring the weights of the WQ parameters [26].

4) Aggregation

The model employs a straightforward additive function to measure the WQI value. The equation is as follows:

$$\mathrm{NSFWQI} = \sum_{i=1}^{n} QiWi$$

5) WQI assessment

NSFWQI value primarily ranges from 0 to 100. Enhancement of the NSFWQI value indicates drinking suitable water. Fathi et al. [90] highlight the five classes of WQI value to assess the WQ, and those are as follows:

i) 91–100 = Very good
ii) 71–90 = Good
iii) 51–70 = Medium
iv) 26–50 = Bad
v) 0–25 = Very bad

IWQI Model

The IWQI was developed by Mukate et al. in [25]. This approach intends to provide a standard paradigm for assessing DWQ. The model mainly contains five-stage, but we discussed it in four stages like earlier.

1) Parameter selection

The model principally selected nine physicochemical items to assess the WQ based on the WQ standard recommendations. The model mainly sets nine things of WQ and selected nine parameters: NO3, SO4, Cl, K, Na, Mg, Ca, TDS, and pH [25].

2) Sub-indexing

The model primarily computes sub-index based on the permissible limit (PL) and desirable limit (DL) provided by the national or international WQ agencies. The model follows some functions to compute the sub-index. The equations are as follows:

$$\mathrm{SI}_1 = 0$$

When the concentration of any WQ elements is under DL or above PL, the value between MPL and DL is considered suitable for drinking; otherwise, it affects WQ. SI_1 will be 0 if the experimental WQ value of the ith element is greater than DL but lower than MPL.

$$\mathrm{SI}_2 = \frac{(\mathrm{DL} - Pi)}{\mathrm{DL}}$$

The above function (SI_2) is employed when the ith item's experiential value (P_i) is lower than the DL but under MPL. The MPL was measured with the help of PL and range.

$$\mathrm{ModifiedPermissibleLimit(MPL)} = [\mathrm{PL} - (20\%\mathrm{Range})]$$

$$\mathrm{Range} = (\mathrm{PL} - \mathrm{DL})$$

At last, SI_3 was calculated when the ith element's observed value (P_i)was greater than the MPL. The formula is as follows:

$$\mathrm{SI}_3 = \frac{(Pi - \mathrm{MPL})}{\mathrm{MPL}}$$

3) Parameter weighting

Parameter weighting is not required for the IWQI model.

4) Aggregation

The model applied a simple additive function to measure the WQI value. The model mainly includes the sum of each sample's sub-indices (SI) to measure the WQI value. The function is as follows:

$$\mathrm{IWQI}i = \sum_{j=1}^{n} \mathrm{SI}_{ij}$$

where SI_{ij} is the sub-index value of the ith sample and jth is the WQ dimensions.

5) WQI assessment

The model outputs are classified into five classes. The classes are as follows:

i) < 1 = Excellent
ii) 1–2 = Good
iii) 2–3 = Marginal
iv) 3–5 = Poor
v) > 5 = Unsuitable

CCME WQI Model

The BCWQI (British Colombian WQI) model was used to build the Canadian Council of Ministers of the Environment (CCME) model [2]. Most researchers applied the CCME model in surface water quality evaluation. However, some researchers adopted the CCME model in assessing DWQ due to its flexibility in WQ selection and ease of use [48, 59]. The model structure is as follows:

1) Parameter selection

The model mainly considered expert opinion in selecting WQ parameters for assessing drinking water suitability. The review observed that the model used 13 physicochemical items for setting DWQ [48, 59].

2) Sub-indexing

The model does not employ any sub-indexing process. It is a significant drawback of this conventional model [2, 25, 91].

3) Parameter weighting

Parameter weighting is not required to get the final WQI value [41, 87, 91, 92].

4) Aggregation

The model applied a different aggregation function than the previous models used in the drinking WQ assessment. The procedure is as follows:

$$\mathrm{CCMEWQI} = 100 - \left[\frac{\sqrt{F_1^2 + F_2^2 + F_3^2}}{1.732}\right]$$

The divisor 1.732 is primarily used as a normalising factor to certify that the WQI value remains within the range of 0 to 100. The determinants (F1, F2, and F3) are definitely in the following way:

I) F1 (scope)

F1 is the % of total variables that will not fulfil the defined goals. The function is as follows:

$$F_1 = \left[\frac{\text{Number of failed parameters}}{\text{Total number of parameters}}\right] \times 100$$

II) F2 (frequency)

The % of independent test results that do not fit the goals is called F2 (failed tests). F2 can be expressed as:

$$F_2 = \left[\frac{\text{Number of failed tests}}{\text{Total number of tests}}\right] \times 100$$

III) F3 (amplitude)

F3 measures how far test values fall short of their goals. The amplitude is computed using an iterative function that uses the following formula to modify the normalised sum of the excursions (NSE) of the test results from the targets to obtain a value between 0 and 100:

$$F_3 = \left[\frac{\mathrm{NSE}}{0.01(\mathrm{NSE}) + 0.01}\right]$$

For a test statistic that falls below the objective value, the excursion is calculated as follows:

$$\mathrm{Excursion}_i = \left[\frac{\text{Failed test value}_i}{\text{Objective}_j}\right] - 1$$

On the other hand, for a test value larger than the objective value, the excursion is calculated as follows:

$$\mathrm{Excursion}_i = \left[\frac{\text{Objective}_j}{\text{failed test value}_i}\right] - 1$$

Based on the excursion value, the NSE is computed. The expression is as follows:

$$NSE = \left[\frac{\sum_{i=1}^{n} \text{excursion}_j}{\text{Total number of test}}\right] - 1$$

5) WQI assessment

The value CCME WQI model primarily ranged from 0 to 100, where 100 shows the best DWQ and 0 shows the poor DWQ. The classes are as follows:

i) 95–100 = Excellent
ii) 80–94 = Good
iii) 65–79 = Fair
iv) 45–64 = Marginal
v) 0–44 = Poor

Discussion

In this study, the researcher critically reviewed the articles and observed that eight primary WQI models are discussed in Table 1. Among eight models, two are modified versions of the ancient WQI or Horton model, and those are PCA-based WQI and WAM-based WQI models. So, the researcher includes it into one and highlights a comparison among the six models. The study also found that the WQI models are adjusted over time and by area to decrease model uncertainty. However, the models face several issues, like eclipsing, parameter selection, weighting, and aggregation problems. In this section, a comparative study has been done concerning the model structure and issues associated with the models.

Model Eclipsing Issues

The incapacity of WQI models to deal with the eclipsing difficulty is one of its shortcomings. Ott [93] has used the word "eclipsing" to describe how the overall result obscures the WQ's essential character [2]. Sub-indexing rules that are not appropriate, item weightings that do not reflect the true relative impacts of variables, and aggregation techniques that are not up to par can all contribute to the eclipsing issue. Several scholars briefly explained eclipsing difficulties [39, 93, 94]. Several investigations, however, have discovered eclipse issues with WQI models [2, 39]. The scholars claim that it is formed due to the aggregation stage's significant usage of mathematical functions.

Many scholars have attempted to prevent eclipsing difficulties, for example, the ANFIS WQI model [28]. The ANFIS WQI model utilises a neural-based fuzzy interface system to reduce eclipsing concerns and improve model reliability [31, 32, 95]. However, when multiple sub-index values from a single determinant are utilised to execute distinct aggregation functions, the final WQI score is altered [2]. However, the ANFIS WQI model helps to minimise the eclipsing concerns, but it also generates uncertainty throughout the aggregation phase [28, 96]. Applying a machine learning (ML) approach may reduce the eclipsing issue [53–55, 57].

Model Uncertainty Concerns

The model uncertainty mainly focuses on how a change in the threshold of the dimension can change the respective values of the sub-index and the final index [85]. Uncertainty is fundamental to every model and can be linked with model-specific parameters [97]. Several scholars identified that uncertainty in the final WQI model indices is related to different sources of the WQI model. The model uncertainty is linked with parameter selection, sub-indexing, and weighting factors [2]. The model uncertainty and eclipses issue is showcased in Table 3. It is noticed that the aggregation function used in the WQI model also includes uncertainty [57]. It is observed that parameter weighting and aggregation function plays a sensitive role in the final index value and model uncertainty. Several scholars are trying to identify the uncertainty issues and apply different statistical analyses to minimise the parameter selection ambiguity in the model, such as discriminant analysis, cluster analysis, component analysis, correlation analysis and machine learning-based functions [57]. Several researchers employed expert advice and environmental importance methodologies to reduce uncertainty in the element assortment and element weighing procedure. Mukate et al. [25] applied PL and DL analysis techniques in sub-indexing to minimise uncertainty. Developing a WQI method should include identifying and measuring ambiguity so that the end WQI scores may be addressed with reliability and utilised to manage water resources and keep them healthy properly.

Parameter Selection

The parameter selection process is a significant step in measuring WQ using the WQI model. It is noticed that the number of WQ parameters significantly varies between models to evaluate drinking water quality. Some WQI models used widespread physicochemical parameters to measure drinking WQ, such as the IWQI model used nine parameters. On the other hand, the ANFIS WQI model used 18 parameters to measure drinking water suitability that is easily accessible. Most DWQI model parameters are determined based on expert opinion and environmental relevance, dependent on WQ data accessibility.

Table 3 A comparative analysis of the selected WQI models

WQI model	Factors contributing to eclipsing	Uncertainty source
PCA- and WAM-based WQI	• The numbers of WQ parameters are not specified • Most of the models are not included biological parameters due to accessibility issues • Parameters are selected based on the literature, expert opinion, and environmental significance	• The aggregation function mainly contributes to uncertainty
EWQI	• EWQI model does not include biological parameters • Parameters weighting has been done with the help of the entropy technique • The complication in sub-index specification	• The main source of uncertainty is in the aggregation function
CCME WQI	• WQ parameters are not specified • Parameter weighting is not required and it is accountable for eclipsing issue	• Application of several aggregation functions, which creates ambiguity in the indexing
NFS WQI	• The parameter selection stage creates eclipsing issue	• Used multiplicative function to reduce the sensitivity of the original function
FIS and ANFIS WQI	• The parameter selection method helps to improve the eclipsing issue	• Advanced mathematical algorithms are the origin of uncertainty
IWQI	• The model only includes general WQ parameters • Biological or toxic elements are not considered • Sub-index calculation is complicated	• The model includes several mathematical functions which can enhance the uncertainty

Some WQI models provide academics with some choice in parameter selection, while others do not. In the case of DWQI, most models like FIS, EWQI, and ANFIS-WQI used user flexibility WQ parameters to assess water quality. However, the scholars need to provide expert opinions and their justification for selecting the specified WQ parameters. It is observed that some scholars use statistical tools to analyse and finalise the WQ parameters, such as the Delphi method. The Delphi technique is well-known for extracting the most acceptable variables from specialists [2, 44]. However, some scholars show that the technique produces data uncertainty and minimises the model's reliability [2]. Therefore, using statistical tools in WQ parameter selection has become popular. In the case of statistical tools, some researchers applied correlation, factor, and PCA analysis to get the best suitable parameters for the DWQ assessment [22, 49]. However, some researchers used local guidelines to evaluate drinking water quality.

Data availability is another crucial consideration while choosing WQ parameters. Some widely used WQI models require comprehensive WQ data to assess DWQ, such as pesticide, toxic, biological, chemical, and biological parameters. However, the availability of comprehensive WQ monitoring data is tricky for developing and underdeveloped countries due to huge costs, improper lab facilities, and labour intensity. In addition, some studies observed improper water quality monitoring as a primary source of error and uncertainty. Numerous developing nations can also not provide modern laboratory facilities for evaluating drinking water due to a deficiency in social support, effective management boards, and economic aid [2].

Recently, researchers have been trying to update the basic WQI models based on the environmental significance of parameters and the availability of new data. For example, the ANFIS model used 18 parameters to adopt new dimensions in the model to evaluate WQ. Some scholars adopted statistical analysis such as PCA, cluster analysis, and spearman rank correlation analysis to assess parameter importance. Researchers and international organisations are also attempting to use biological and toxic characteristics to determine the appropriateness of drinking water. E. coli, for example, is a better parameter to measure DWQ than F. Coli, according to the WHO and EU. They recommended that future WQI models include the parameter for assessing drinking and bathing WQ [2, 98].

Sub-indexing

Sub-indexing is a crucial part of the WQI model and has a relative influence on the final index [2]. However, some WQI models have not included the sub-indexing stage to assess DWQ, such as the SRDD, CCME, Smith Index, and Oregon index. Some WQI models directly used parameters measure value as sub-index values, and some used complex mathematical formulas to measure sub-indices, such as NSF and FIS models [2]. However, expert opinion for calculating the sub-index rule for components was also noticed [2, 57]. However, a significant concern is required in the sub-indexing calculation so that the measurement minimises the uncertainty. When designing approaches for obtaining sub-index values, it is essential to remember that the obtained values should not obscure the parameter's significance. One of the significant drawbacks of sub-indexing systems is that they obscure the primary information about WQ. The drawback mainly leads to eclipsing and uncertainty issues. Several scholars highlight that local guidelines can be utilised to build suitable sub-indices; the values should be compared with the overseas guideline values to enhance the reliability of the WQI [2]. However, in recent decades, some mathematical functions are developed to deal with such issues such as Sensitivity analysis, machine learning (ML) approach, artificial intelligence (AI), and stimulation technique [37, 53, 54, 57]

Parameter Weighting

It is a crucial stage in the WQI model since it determines a WQ parameter's relative effect on the final model [66, 87, 99]. It was observed that most of the WQI models use weighting for evaluating DWQ. Most scholars apply unequal weighting systems in the models because unequal weighting can make a difference among the influence of different parameters [2, 53–55, 57]. Many WQI models adopt expert opinion (NSF), entropy weight (EWQI), and permissible limit (i.e., IWQI) in the parameter weighting process. The expert panels mainly provide weighting values based on the environmental significance, the purpose of the study, and standard guidelines. For instance, dissolved oxygen has several weights in different models, such as 4 (Horton), 0.17 (NSF), 0.18 (SRDD), and 0.2 (House). It is also noticed that the same WQ parameter weight varies in different models, making it challenging to select the appropriate weight for the parameters. Recent studies highlight that the AHP method helps determine the parameter's importance [2, 37] and therefore helps minimise the uncertainty resulting from the unsuitable weighting of WQ items. However, recently developed ML and AI-based models (such as RMS and AM) also help reduce the weighting process's uncertainty [57].

Aggregation Function

Scholars in measuring DWQ have used different types of aggregation functions. Multiplicative and additive functions are the most popular aggregation technique in the WQI models [2]. Some research shows that these two functions are involved in the eclipsing issues and general uncertainty in the WQI models [37]. Recent studies show that scholars are trying to modify the functions used in the WQI models to reduce model uncertainty. For example, the IWQI model used permissible and desirable limits for measuring the sub-indices of WQ parameters, and EWQI used the entropy method in the sub-indices calculation to reduce model uncertainty [2, 12, 25]. Some scholars used computer-based advanced mathematical modelling (such as ANN, FIS, ML, and AI-based models) to improve the reliability of WQI models [2, 81, 84].

WQI Classification

Another critical issue with the WQI model is that the classification range differs in different WQI models. So, the WQ classes exhibit a wide range of differences. As a result, determining the most realistic WQ situations is challenging. As the WQI model steadily gains more significant popularity, worries about these issues are becoming increasingly widespread. This uncertainty is reflected in the weakness of the WQI model's aggregation process. There is a great deal of confusion in the proper categorisation of water quality since various approaches provide a variety of interpretations for the same water attributes. As "metaphoring difficulties" of categorisation, these issues may be solved [56]. Consequently, specific requirements for developing an appropriate WQI model for analysing real-world DWQ scenarios are critical [2, 53–55, 57, 87]. Following that, a reliable WQI model is required to measure DWQ accurately. To deal with this issue, M. G. Uddin et al. [56] developed a universal classification scheme for assessing water quality.

Conclusion

Due to their relative straightforwardness and instantly relevant output, WQI has become a standard method for WQ evaluation; nevertheless, various modified WQI models have also been reported. This study aimed to examine the model structure and statistical methodologies employed in WQI models. The study concludes that most models had roughly parallel systems; the more delicate niceties of the four fundamental stages differed significantly. The review also brought up the difficulties of eclipsing and ambiguity that arise throughout the model construction process. Three significant findings from the study are as follows:

- It is observed that most of the WQI models include five stages to measure DWQ. Those are (1) parameter selection, (2) sub-indexing, (3) parameter weighting, (4) aggregation function, and (5) classification of WQI value. However, it will be four for the unweighted-based models, as the unweighted models did not consider the parameter weighting process. The first stage involves the selection of important WQ parameters to measure DWQ. Most of the models used some common WQ parameters to measure DWQ. However, there is a variation in parameter selection according to the standard guideline, expert opinion, data availability, statistical technique, and environmental significance. The sub-indexing stage mainly includes the determination of parameter sub-indices. There is also a wide range of techniques observed to calculate sub-indices. Some model directly adopts the WQ parameters value, and some use statistical techniques to measure the sub-indices. The third stage includes the determination of parameter weighting using different statistical tools and expert opinion. The fourth stage, or aggregation function, involves the computation of overall WQI to get a single value. The last stage involved the classification of the WQI value. Two common aggregation functions were observed: multiplicative and additional functions. It is noticed that most models use the four stages mentioned in WQI measurement. However, the models are quite region-specific and vary according to the types of water body assessment (i.e., groundwater/estuary/river/coastal).

- A significant variation is observed in the quantity and type of WQ parameters incorporated in WQI models, the relative weighting of the parameters, and sub-indices calculation. As a result, there is minimal consistency among the models, making it difficult to compare the models applied in the different study areas. However, WQI models may become more appealing instruments for water quality evaluation if their structure and methods are simplified by including global standard values (i.e., EU WFD and WHO). The modification of the WQI models can help improve the usability of the models, such as the inclusion of new parameters (i.e., microbiological, toxic, and pesticide parameters) and the adoption of the new method (i.e., ANN, ML, AI, and FIS) to reduce uncertainty in the model. Recent studies need to take care of the model selection and model-appropriate application.
- The most critical factors impacting model outputs' accuracy are eclipsing and uncertainty. WQI models at all four levels can help here. Regarding parameter selection, formulation of sub-indexing methods, determining suitable weightings, and model development have depended primarily on expert panel judgements. However, relying on individual opinions may be involved in the creation of uncertainty in the model. Computer-based methods such as FIS and ANFIS have been applied to minimise the uncertainty in the final stage, and mathematical approaches such as PCA, FA, correlation, ML, and AI have been used to improve the selection of WQ items and their parameter weighting. The techniques should be applied to minimise the uncertainty of the WQI model. At last, it can be said that model uncertainty should be examined for every WQI model used to evaluate DWQ.

The study provides several theoretical and practical implications for practitioners and scholars concerning drinking water quality assessment. First, the study provides a brief review concerning the parameter selection, sub-indexing, parameter weighting, and aggregation function used in the drinking water quality assessment. Second, the study briefly discusses the recent developments of WQI models, such as ML and AI-based models, to deal with the uncertainty issue. However, the study has several limitations, such as only considering articles mainly developed based on drinking water quality assessment. The study also considered a small time frame which may exclude some exciting and well-organised models concerning water quality assessment.

Acknowledgements The authors are thankful to Vidyasagar University for providing good research environment.

Author Contribution The study's conception and design were aided by all of the authors. The first author [Atanu Manna] conducted the literature search and data presentation in this review article, which was drafted and critically revised by the corresponding author [Dr. Debasish Biswas].

Funding The authors state that they did not receive any funding, grants, or other forms of support in the development of this paper.

Data Availability It is a review-based article and the gathered secondary information is highlighted in the supplementary file.

Declarations

Competing interests The authors declare no competing interests.

Ethical Approval and Consent to Participate The authors strictly adhere to all ethical considerations during the literature search and data presentation to conduct this study.

Consent for Publication Not applicable.

Competing Interest The authors declare no competing interests.

References

1. Damo R, Icka P (2013) Evaluation of water quality index for drinking water. Pol J Environ Stud 22(4):1045–1051. https://www.researchgate.net/profile/Pirro-Icka/publication/287957321_Evaluation_of_Water_Quality_Index_for_Drinking_Water/links/5923fd63aca27295a8aad7c1/Evaluation-of-Water-Quality-Index-for-Drinking-Water.pdf. Accessed 6 Nov 2021
2. Uddin MG, Nash S, Olbert AI (2021) A review of water quality index models and their use for assessing surface water quality. Ecol Indic 122:107218. https://doi.org/10.1016/j.ecolind.2020.107218
3. Uddin MG, Moniruzzaman M, Quader MA, Hasan MA (2018) Spatial variability in the distribution of trace metals in groundwater around the Rooppur nuclear power plant in Ishwardi, Bangladesh. Groundw Sustain Dev 7:220–231. https://doi.org/10.1016/j.gsd.2018.06.002
4. Egbueri JC, Ameh PD, Unigwe CO (2020) Integrating entropy-weighted water quality index and multiple pollution indices towards a better understanding of drinking water quality in Ojoto area, SE Nigeria. Sci African 10:e00644. https://doi.org/10.1016/j.sciaf.2020.e00644
5. Ortega DJP, Pérez DA, Américo JHP, de Carvalho SL, Segovia JA (2016) Development of index of resilience for surface water in watersheds. J Urban Environ Eng 10(1):72–82. https://doi.org/10.4090/juee.2016.v10n1.007282
6. Alcamo J (2019) Water quality and its interlinkages with the sustainable development goals. Curr Opin Environ Sustain 36:126–140. https://doi.org/10.1016/j.cosust.2018.11.005

7. Li P, Wu J (2019) Drinking water quality and public health. Exposure and Health 11(2):73–79. https://doi.org/10.1007/s12403-019-00299-8
8. Guo X, Zhang XX, Yue HC (2018) Evaluation of hierarchically weighted principal component analysis for water quality management at Jiaozuo mine. Int Biodeterior Biodegradation 128:182–185. https://doi.org/10.1016/j.ibiod.2017.11.012
9. Ighalo JO, Adeniyi AG (2020) A comprehensive review of water quality monitoring and assessment in Nigeria. Chemosphere, 127569. https://doi.org/10.30564/jees.v3i1.2900
10. Motlagh AM, Yang Z, Saba H (2020) Groundwater quality. Water Environ Res 92(10):1649–1658. https://doi.org/10.1002/wer.1412
11. Zhang Q, Xu P, Qian H (2020) Groundwater quality assessment using improved water quality index (WQI) and human health risk (HHR) evaluation in a semi-arid region of northwest China. Exposure and Health 12(3):487–500. https://doi.org/10.1007/s12403-020-00345-w
12. Egbueri JC, Ezugwu CK, Ameh PD, Unigwe CO, Ayejoto DA (2020) Appraising drinking water quality in Ikem rural area (Nigeria) based on chemometrics and multiple indexical methods. Environ Monit Assess 192(5). https://doi.org/10.1007/s10661-020-08277-3
13. Fu B, Merritt WS, Croke BFW, Weber TR, Jakeman AJ (2019) A review of catchment-scale water quality and erosion models and a synthesis of future prospects. Environ Model Softw 114:75–97. https://doi.org/10.1016/j.envsoft.2018.12.008
14. Hui T, Xiujuan L, Qifa S, Qiang L, Zhuang K, Yan G (2020) Evaluation of drinking water quality using the water quality index (WQI), the synthetic pollution index (SPI) and geospatial tools in Lianhuashan District, China. Pol J Environ Stud 30(1):141–153. https://doi.org/10.15244/pjoes/120765
15. Lumb A, Sharma TC, Bibeault J-F (2011) A review of genesis and evolution of water quality index (WQI) and some future directions. Water Qual Expo Health 3(1):11–24. https://doi.org/10.1007/s12403-011-0040-0
16. Solangi GS, Siyal AA, Babar MM, Siyal P (2019a) Application of water quality index, synthetic pollution index, and geospatial tools for the assessment of drinking water quality in the Indus Delta, Pakistan. Environ Monit Assess, 191(12). https://doi.org/10.1007/s10661-019-7861-x
17. Abba SI, Hadi SJ, Sammen SS, Salih SQ, Abdulkadir RA, Pham QB, Yaseen ZM (2020) Evolutionary computational intelligence algorithm coupled with self-tuning predictive model for water quality index determination. J Hydrol 587:124974. https://doi.org/10.1016/j.jhydrol.2020.124974
18. Singh B, Sihag P, Singh VP, Sepahvand A, Singh K (2021) Soft computing technique-based prediction of water quality index. Water Supply. https://doi.org/10.2166/ws.2021.157
19. Tung TM, Yaseen ZM (2021) Deep learning for prediction of water quality index classification: tropical catchment environmental assessment. Nat Resour Res 30:4235–4254. https://doi.org/10.1007/s11053-021-09922-5
20. Banda TD, Kumarasamy M (2020) Development of a universal water quality index (UWQI) for South African river catchments. Water 12(6):1534. https://doi.org/10.3390/W12061534
21. Mamun M, An KG (2021) Application of multivariate statistical techniques and water quality index for the assessment of water quality and apportionment of pollution sources in the Yeongsan river, South Korea. Int J Environ Res Public Health 18(16):8268. https://doi.org/10.3390/ijerph18168268
22. Tripathi M, Singal SK (2019) Use of principal component analysis for parameter selection for development of a novel water quality index: a case study of river Ganga India. Ecol Ind 96:430–436. https://doi.org/10.1016/j.ecolind.2018.09.025
23. Abbasnia A, Yousefi N, Mahvi AH, Nabizadeh R, Radfard M, Yousefi M, Alimohammadi M (2019) Evaluation of groundwater quality using water quality index and its suitability for assessing water for drinking and irrigation purposes: case study of Sistan and Baluchistan province (Iran). Hum Ecol Risk Assess Int J 25(4):988–1005. https://doi.org/10.1080/10807039.2018.1458596
24. Jha MK, Shekhar A, Jenifer MA (2020) Assessing groundwater quality for drinking water supply using hybrid fuzzy-GIS-based water quality index. Water Res 179:115867. https://doi.org/10.1016/j.watres.2020.115867
25. Mukate S, Wagh V, Panaskar D, Jacobs JA, Sawant A (2019) Development of new integrated water quality index (IWQI) model to evaluate the drinking suitability of water. Ecol Ind 101:348–354. https://doi.org/10.1016/j.ecolind.2019.01.034
26. Al-Hamdany NAS, Al-Shaker YMS, Al-Saffawi AYT (2020) Water quality assessment using the NSFWQI model for drinking and domestic purposes: a case study of groundwater on the left side of Mosul city, Iraq. Plant Archives 20(1):3079–3085. http://www.plantarchives.org/20-1/3079-3085(6183).pdf. Accessed 6 Nov 2021
27. Bouteraa O, Mebarki A, Bouaicha F, Nouaceur Z, Laignel B (2019) Groundwater quality assessment using multivariate analysis, geostatistical modeling, and water quality index (WQI): a case of study in the Boumerzoug-El Khroub valley of Northeast Algeria. Acta Geochimica 38(6):796–814. https://doi.org/10.1007/s11631-019-00329-x
28. RadFard M, Seif M, GhazizadehHashemi AH, Zarei A, Saghi MH, Shalyari N, Morovati R, Heidarinejad Z, Samaei MR (2019) Protocol for the estimation of drinking water quality index (DWQI) in water resources: artificial neural network (ANFIS)and Arc-Gis. MethodsX 6:1021–1029. https://doi.org/10.1016/j.mex.2019.04.027
29. Solangi GS, Siyal AA, Babar MM, Siyal P (2019) Evaluation of drinking water quality using the water quality index (WQI), the synthetic pollution index (SPI) and geospatial tools in Thatta district, Pakistan. Desalination Water Treat 160:202–213. https://doi.org/10.5004/dwt.2019.24241
30. Solangi GS, Siyal AA, Babar MM, Siyal P (2020) Groundwater quality evaluation using the water quality index (WQI), the synthetic pollution index (SPI), and geospatial tools: a case study of Sujawal district, Pakistan. Human Ecol Risk Assess 26(6):1529–1549. https://doi.org/10.1080/10807039.2019.1588099
31. Tiwari AK, Singh AK, Mahato MK (2018) Assessment of groundwater quality of Pratapgarh district in India for suitability of drinking purpose using water quality index (WQI) and GIS technique. Sustain Water Resour Manag 4(3):601–616. https://doi.org/10.1007/s40899-017-0144-1
32. Tiwari S, Babbar R, Kaur G (2018). Performance evaluation of two ANFIS models for predicting water quality Index of River Satluj (India). Adv Civil Eng 2018. https://doi.org/10.1155/2018/8971079
33. Seifi A, Dehghani M, Singh VP (2020) Uncertainty analysis of water quality index (WQI) for groundwater quality evaluation: application of Monte-Carlo method for weight allocation. Ecol Indic 117:106653. https://doi.org/10.1016/j.ecolind.2020.106653
34. Dutta N, Thakur BK, Nurujjaman M, Debnath K, Bal DP (2022) An assessment of the water quality index (WQI) of drinking water in the Eastern Himalayas of South Sikkim India. Groundw Sustain Dev 17:100735. https://doi.org/10.1016/j.gsd.2022.100735
35. Verma P, Singh PK, Sinha RR, Tiwari AK (2020) Assessment of groundwater quality status by using water quality index (WQI) and geographic information system (GIS) approaches: a case study of the Bokaro district, India. Appl Water Sci 10(1):1–16. https://doi.org/10.1007/s13201-019-1088-4
36. Gorai AK, Hasni SA, Iqbal J (2016) Prediction of ground water quality index to assess suitability for drinking purposes using fuzzy rule-based approach. Appl Water Sci 6(4):393–405. https://doi.org/10.1007/s13201-014-0241-3
37. Akhtar N, Ishak MIS, Ahmad MI, Umar K, MdYusuff MS, Anees MT, Qadir A, Ali Almanasir YK (2021) Modification of the water quality index (WQI) process for simple calculation using

the multi-criteria decision-making (MCDM) method: a review. Water 13(7):905. https://doi.org/10.3390/w13070905
38. Aljanabi ZZ, Al-Obaidy A-HMJ, Hassan FM (2021) A brief review of water quality indices and their applications. IOP Conf Ser: Earth Environ Sci 779(1):12088. https://doi.org/10.1088/1755-1315/779/1/012088
39. Soumaila KI, Niandou AS, Naimi M, Mohamed C, Schimmel K, Luster-Teasley S, Sheick NN (2019) A systematic review and meta-analysis of water quality indices. J Agric Sci Technol B B(9):1–14. https://doi.org/10.17265/2161-6264/2019.01.001
40. Abed BS, Farhan A-R, Ismail AH, Al Aani S (2021) Water quality index toward a reliable assessment for water supply uses: a novel approach. Int J Environ Sci Technol, 1–14. https://doi.org/10.1007/s13762-021-03338-7
41. Wagh VM, Mukate SV, Panaskar DB, Muley AA, Sahu UL (2019) Study of groundwater hydrochemistry and drinking suitability through Water Quality Index (WQI) modelling in Kadava river basin, India. SN Applied Sciences 1(10):1–16. https://doi.org/10.1007/s42452-019-1268-8
42. Gupta S, Gupta SK (2021) A critical review on water quality index tool: genesis, evolution and future directions. Eco Inform 63:101–299. https://doi.org/10.1016/j.ecoinf.2021.101299
43. Hui T, Jizhong D, Qifa S, Yan G, Zhuang K, Hongtao J (2021) Evaluation of shallow groundwater for drinking purpose based on water quality index and synthetic pollution index in Changchun New District, China. Environ Forensic 22(1–2):189–204. https://doi.org/10.1080/15275922.2020.1834024
44. Ewaid SH, Abed SA, Al-Ansari N, Salih RM (2020) Development and evaluation of a water quality index for the Iraqi rivers. Hydrology 7(3):67. https://doi.org/10.3390/HYDROLOGY7030067
45. Ukah BU, Ameh PD, Egbueri JC, Unigwe CO, Ubido OE (2020) Impact of effluent-derived heavy metals on the groundwater quality in Ajao industrial area, Nigeria: an assessment using entropy water quality index (EWQI). Int J Energy Water Resour, 1–14. https://doi.org/10.1007/s42108-020-00058-5
46. Tiri A, Belkhiri L, Mouni L (2018) Evaluation of surface water quality for drinking purposes using fuzzy inference system. Groundw Sustain Dev 6:235–244. https://doi.org/10.1016/j.gsd.2018.01.006
47. Alver A (2019) Evaluation of conventional drinking water treatment plant efficiency according to water quality index and health risk assessment. Environ Sci Pollut Res 26(26):27225–27238. https://doi.org/10.1007/s11356-019-05801-y
48. Saw S, Mahato JK, Singh PK (2021). Suitability Evaluation of CCME-WQI and GWQI for the modeling of groundwater and human health risk assessment of heavy metals - Eastern India. Res Square, 0–24. https://doi.org/10.21203/rs.3.rs-1000020/v1
49. Patil VBB, Pinto SM, Govindaraju T, Hebbalu VS, Bhat V, Kannanur LN (2020) Multivariate statistics and water quality index (WQI) approach for geochemical assessment of groundwater quality—a case study of Kanavi Halla Sub-Basin, Belagavi, India. Environ Geochem Health 42(9):2667–2684. https://doi.org/10.1007/s10653-019-00500-6
50. Nath BK, Chaliha C, Bhuyan B, Kalita E, Baruah DC, Bhagabati AK (2018) GIS mapping-based impact assessment of groundwater contamination by arsenic and other heavy metal contaminants in the Brahmaputra River valley: a water quality assessment study. J Clean Prod 201:1001–1011. https://doi.org/10.1007/s10462-021-10007-1
51. Sutadian AD, Muttil N, Yilmaz AG, Perera BJC (2016) Development of river water quality indices—a review. Environ Monit Assess 188(1):58. https://doi.org/10.1007/s10661-015-5050-0
52. Swamee PK, Tyagi A (2007) Improved method for aggregation of water quality subindices. J Environ Eng 133(2):220–225. https://doi.org/10.1061/(ASCE)0733-9372(2007)133:2(220)
53. Uddin G, Nash S, Olbert AI (2022) Optimisation of parameters in a water quality index model using principal component analysis. Proc 39th IAHR World Congr 19:5739–5744. https://doi.org/10.3850/IAHR-39WC2521711920221326
54. Uddin MG, Nash S, Rahman A, Olbert AI (2022) A comprehensive method for improvement of water quality index (WQI) models for coastal water quality assessment. Water Research 219:118532. https://doi.org/10.1016/j.watres.2022.118532
55. Uddin MG, Nash S, Rahman A, Olbert AI (2023) Assessing optimisation techniques for improving water quality model. J Clean Prod 385:135671. https://doi.org/10.1016/j.jclepro.2022.135671
56. Uddin MG, Nash S, Rahman A, Olbert AI (2023) Performance analysis of the water quality index model for predicting water state using machine learning techniques. Process Saf Environ Prot 169:808–828. https://doi.org/10.1016/j.psep.2022.11.073
57. Uddin MG, Nash S, Rahman A, Olbert AI (2023) A novel approach for estimating and predicting uncertainty in water quality index model using machine learning approaches. Water Res 229:119422. https://doi.org/10.1016/j.watres.2022.119422
58. Islam M, Mostafa MG (2021) Development of an integrated irrigation water quality index (IIWQIndex) model. Water Supply 22(2):2322–2337. https://doi.org/10.2166/ws.2021.378
59. Wagh VM, Panaskar DB, Muley AA, Mukate SV (2017) Groundwater suitability evaluation by CCME WQI model for Kadava River Basin, Nashik, Maharashtra, India. Model Earth Syst Environ 3(2):557–565. https://doi.org/10.1007/s40808-017-0316-x
60. Elbeltagi A, Pande CB, Kouadri S, Islam ARM (2022) Applications of various data-driven models for the prediction of groundwater quality index in the Akot basin, Maharashtra India. Environ Sci Pollut Res 29(12):17591–17605. https://doi.org/10.1007/s11356-021-17064-7
61. Hossain M, Patra PK (2020) Water pollution index–a new integrated approach to rank water quality. Ecol Indic 117:106668. https://doi.org/10.1016/j.ecolind.2020.106668
62. Barbosa Filho J, de Oliveira IB (2021) Development of a groundwater quality index: GWQI, for the aquifers of the state of Bahia, Brazil using multivariable analyses. Sci Rep 11(1):1–22. https://doi.org/10.1038/s41598-021-95912-9
63. Gradilla-Hernández MS, de Anda J, Garcia-Gonzalez A, Montes CY, Barrios-Piña H, Ruiz-Palomino P, Díaz-Vázquez D (2020) Assessment of the water quality of a subtropical lake using the NSF-WQI and a newly proposed ecosystem specific water quality index. Environ Monit Assess 192(5):1–19. https://doi.org/10.1007/s10661-020-08265-7
64. Najafzadeh M, Homaei F, Farhadi H (2021) Reliability assessment of water quality index based on guidelines of national sanitation foundation in natural streams: integration of remote sensing and data-driven models. Artif Intell Rev 56(4):4619–4651. https://doi.org/10.1007/s10462-021-10007-1
65. Zotou I, Tsihrintzis VA, Gikas GD (2019) Performance of seven water quality indices (WQIs) in a Mediterranean River. Environ Monit Assess 191(8):1–14
66. Banda, Kumarasamy MA (2020) Review of the existing water quality indices (WQIs). J Phys Opt 39(2):1–19. https://www.researchgate.net/profile/Talent-Banda/publication/343430598_A_Review_of_the_Existing_Water_Quality_Indices_WQIs/links/5f299e76299bf13404a22edc/A-Review-of-the-Existing-Water-Quality-Indices-WQIs.pdf. Accessed 11 Nov 2021
67. Kamboj V, Kamboj N, Bisht A (2020) An overview of water quality indices as promising tools for assessing the quality of water resources. Agro Environ Media - Agric Environ Sci Acad pp. 188–214. https://doi.org/10.26832/aesa-2020-aepm-013
68. Haider H, Ghumman AR, Al-Salamah IS, Thabit H (2020) Assessment framework for natural groundwater contamination in arid regions: development of indices and wells ranking system using fuzzy VIKOR method. Water 12(2):423. https://doi.org/10.3390/w12020423

69. Pak HY, Chuah CJ, Tan ML, Yong EL, Snyder SA (2021) A framework for assessing the adequacy of water quality index–quantifying parameter sensitivity and uncertainties in missing values distribution. Sci Total Environ 751(10):141982. https://doi.org/10.1016/j.scitotenv.2020.141982
70. Smith DG (1990) A better water quality indexing system for rivers and streams. Water Res 24(10):1237–1244. https://doi.org/10.1016/0043-1354(90)90047-A
71. Shah KA, Joshi GS (2017) Evaluation of water quality index for River Sabarmati, Gujarat, India. Appl Water Sci 7(3):1349–1358. https://doi.org/10.1007/s13201-015-0318-7
72. Said A, Stevens DK, Sehlke G (2004) An innovative index for evaluating water quality in streams. Environ Manage 34(3):406–414. https://doi.org/10.1007/s00267-004-0210-y
73. Adimalla N, Qian H, Li P (2019) Entropy water quality index and probabilistic health risk assessment from geochemistry of groundwaters in hard rock terrain of Nanganur County, South India. Geochemistry 80(4):125544. https://doi.org/10.1016/j.chemer.2019.125544
74. Long Y, Yang Y, Lei X, Tian Y, Li Y (2019) Integrated assessment method of emergency plan for sudden water pollution accidents based on improved TOPSIS, Shannon entropy and a coordinated development degree model. Sustainability 11(2):510. https://doi.org/10.1007/s12403-019-00299-8
75. Yang Z, Wang Y (2020) The cloud model based stochastic multi-criteria decision making technology for river health assessment under multiple uncertainties. J Hydrol 581:124437. https://doi.org/10.1016/j.jhydrol.2019.124437
76. Hanoon MS, Ahmed AN, Fai CM, Birima AH, Razzaq A, Sherif M, Sefelnasr A, El-Shafie A (2021) Application of artificial intelligence models for modeling water quality in groundwater: comprehensive review, evaluation and future trends. Water Air Soil Pollut 232(10):1–41. https://doi.org/10.1007/s11270-021-05311-z
77. Chanapathi T, Thatikonda S (2019) Fuzzy-based regional water quality index for surface water quality assessment. J Hazard Toxic Radioact Waste 23(4):4019010. https://doi.org/10.1061/(asce)hz.2153-5515.0000443
78. Hamdan ANA, Al Saad ZAA, Abu-Alhail S (2021) Fuzzy system modelling to assess water quality for irrigation purposes. J Water Land Dev 50:98–107. https://doi.org/10.24425/jwld.2021.138165
79. Hue NH, Thanh NH (2020) Surface water quality analysis using Fuzzy logic approach: a case of inter-provincial irrigation network in Vietnam. IOP Conf Ser: Earth Environ Sci 527(1):12017. https://doi.org/10.1088/1755-1315/527/1/012017
80. Lindang HU, Tarmudi ZH, Jawan A (2017) Assessing water quality index in river basin: Fuzzy inference system approach. Malays J Geosci 1(1):27–31. https://doi.org/10.1007/s12403-019-00299-8
81. Sharifi H, Roozbahani A, Shahdany SMH (2021) Evaluating the performance of agricultural water distribution systems using FIS, ANN and ANFIS intelligent models. Water Resour Manage 35(6):1797–1816
82. Selvaraj A, Saravanan S, Jennifer JJ (2020) Mamdani fuzzy based decision support system for prediction of groundwater quality: an application of soft computing in water resources. Environ Sci Pollut Res 27(20):25535–25552. https://doi.org/10.1007/s11356-020-08803-3
83. Hajji S, Yahyaoui N, Bousnina S, Ben Brahim F, Allouche N, Faiedh H, Bouri S, Hachicha W, Aljuaid AM (2021) Using a Mamdani Fuzzy Inference System Model (MFISM) for ranking groundwater quality in an agri-environmental context: case of the Hammamet-Nabeul shallow aquifer (Tunisia). Water 13(18):2507. https://doi.org/10.3390/w13182507
84. Kambalimath S, Deka PC (2020) A basic review of fuzzy logic applications in hydrology and water resources. Appl Water Sci 10(8):191. https://doi.org/10.1007/s13201-020-01276-2
85. Islam AR, Al Mamun A, Rahman MM, Zahid A (2020) Simultaneous comparison of modified-integrated water quality and entropy weighted indices: implication for safe drinking water in the coastal region of Bangladesh. Ecol Indic 113:106229. https://doi.org/10.1016/j.ecolind.2020.106229
86. Rabeiy RES (2018) Assessment and modeling of groundwater quality using WQI and GIS in Upper Egypt area. Environ Sci Pollut Res 25(31):30808–30817. https://doi.org/10.1007/s11356-017-8617-1
87. Uddin MG, Olbert AI, Nash S (2020) Assessment of water quality using water quality index (WQI). CERI 2020 Proc 85:966–982
88. Talat RA, Al-Assaf AYR, Al-Saffawi AYT (2019) Valuation of water quality for drinking and domestic purposes using WQI : a case study for groundwater of Al-Gameaa and Al-Zeraee quarters in Mosul city/Iraq. J Phys: Conf Ser 1294(7). https://doi.org/10.1088/1742-6596/1294/7/072011
89. Kachroud M, Trolard F, Kefi M, Jebari S, Bourrié G (2019) Water quality indices: challenges and application limits in the literature. Water 11(2):361. https://doi.org/10.3390/w11020361
90. Fathi E, Zamani-Ahmadmahmoodi R, Zare-Bidaki R (2018) Water quality evaluation using water quality index and multivariate methods, Beheshtabad River, Iran. Appl Water Sci 8(7):1–6. https://doi.org/10.1007/s13201-018-0859-7
91. Feng Y, Bao Q, Chenglin L, Bowen W, Zhang Y (2018) Introducing biological indicators into CCME WQI using variable fuzzy set method. Water Resour Manage 32(8):2901–2915. https://doi.org/10.1007/s13201-018-0859-7
92. Gikas GD, Sylaios GK, Tsihrintzis VA, Konstantinou IK, Albanis T, Boskidis I (2020) Comparative evaluation of river chemical status based on WFD methodology and CCME water quality index. Sci Total Environ 745:140849. https://doi.org/10.1016/j.scitotenv.2020.140849
93. Ott WR (1978) Environmental indices: theory and practice. Ann Arbor Science Publishers, Inc., Ann Arbor, MI. https://www.osti.gov/biblio/6681348. Accessed 13 Nov 2021
94 Ismail AH, Robescu D (2019) Assessment of water quality of the Danube river using water quality indices technique. Environ Eng Manag J 18(8):1727–1737. https://doi.org/10.30638/eemj.2019.163
95. Khan Y, Chai SS (2017) Ensemble of ANN and ANFIS for water quality prediction and analysis-a data driven approach. J Telecommun Electron Comput Eng (JTEC) 9(29):117–122
96. Patki VK, Jahagirdar S, Patil YM, Karale R, Nadagouda A (2021). Prediction of water quality in municipal distribution system. Materials Today: Proceedings. https://doi.org/10.1016/j.matpr.2021.02.826
97. Volodina V, Challenor P (2021) The importance of uncertainty quantification in model reproducibility. Phil Trans R Soc A 379(2197):20200071. https://doi.org/10.1098/rsta.2020.0071
98. Rodrigues C, Cunha MÂ (2017) Assessment of the microbiological quality of recreational waters: indicators and methods. Euro-Mediterr J Environ Integr 2(1):1–18. https://doi.org/10.1007/s41207-017-0035-8
99. Heiß L, Bouchaou L, Tadoumant S, Reichert B (2020) Index-based groundwater vulnerability and water quality assessment in the arid region of Tata city (Morocco). Groundw Sustain Dev 10:100344. https://doi.org/10.1016/j.gsd.2020.100344